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Set 16: Customer Demographics Analysis

Data Visualization using R

Aim

To analyse customer demographics using a bar chart, a pie chart, a table and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the customer data as a data frame and group ages with `cut()`.
3. Question 1: draw the age bar chart with `geom_bar()`.
4. Question 2: draw the gender pie chart with `geom_col()` and `coord_polar("y")`.
5. Question 3: make the table with `tableGrob()`.
6. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
cust <- data.frame(Customer_ID = 1:3, Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"), Income = c(50000, 60000, 75000))
print(cust)
cust$Age_Group <- cut(cust$Age, c(20, 30, 40, 50), right = FALSE,
  labels = c("20-29", "30-39", "40-49"))
gender <- as.data.frame(table(cust$Gender)); names(gender) <- c("Gender", "Count")
print(gender)

# 1. Bar chart - age distribution
p1 <- ggplot(cust, aes(Age_Group)) + geom_bar(fill = "steelblue") +
  labs(title = "Customer Age Distribution", x = "Age Group", y = "Customers")

# 2. Pie chart - gender
p2 <- ggplot(gender, aes("", Count, fill = Gender)) + geom_col() + coord_polar("y") +
  geom_text(aes(label = Count), position = position_stack(vjust = 0.5)) +
  labs(title = "Customers by Gender") + theme_void()

# 3. Table
tb <- tableGrob(cust[, 1:4], rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Customer Dashboard")

ggsave("set16_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set16_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set16_q3.png", arrangeGrob(tb, top = "Customer Demographics Table"), width = 5, height = 2, bg = "white")
ggsave("set16_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

```
Customer_ID Age Gender Income
1      1  28 Female 50000
2      2  35  Male 60000
3      3  42 Female 75000
Gender Count
1 Female    2
2 Male      1
```

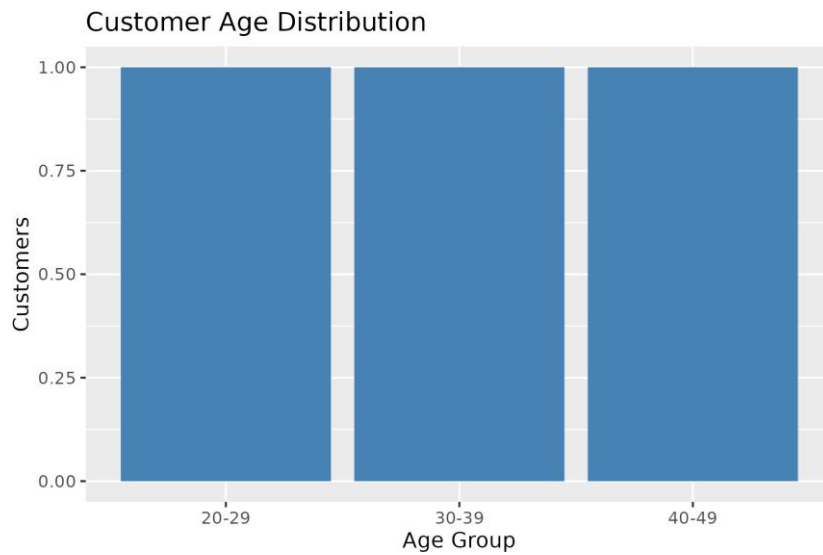


Figure 16.1: Q1 - Age distribution bar chart

Customers by Gender

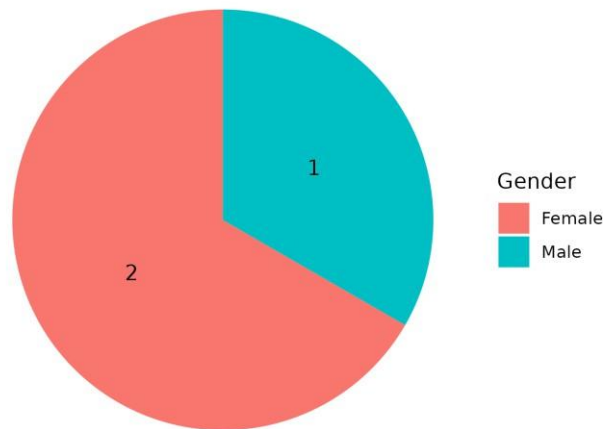
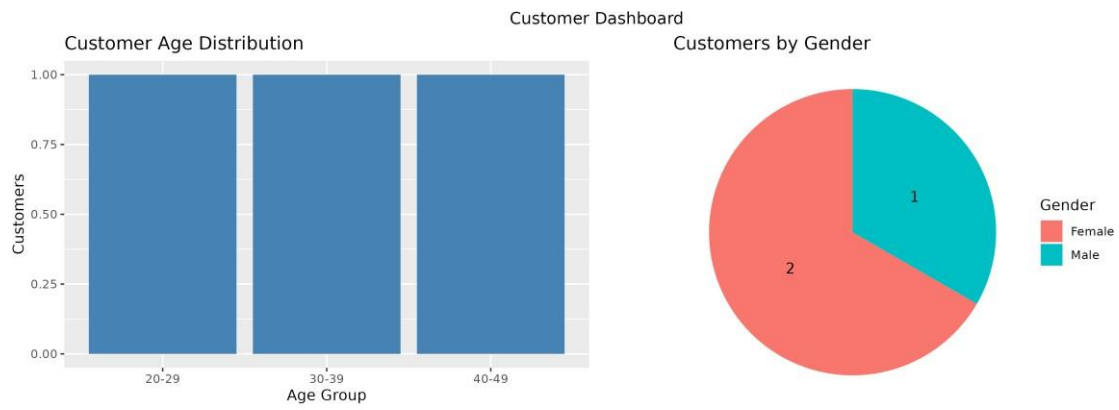


Figure 16.2: Q2 - Gender pie chart

Customer Demographics Table

Customer_ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Figure 16.3: Q3 - Customer table



Customer_ID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Figure 16.4: Q4 - Dashboard

Result

- There is one customer in each age group (20-29, 30-39, 40-49).
- Two customers are female and one is male.
- All four parts were completed successfully in R.

